

CSE 4631

Digital Signal Processing

Week 4

Requirements of Linearity

- A system is called *linear* if it has two mathematical properties
 - Homogeneity / multiplicative
 - Additivity

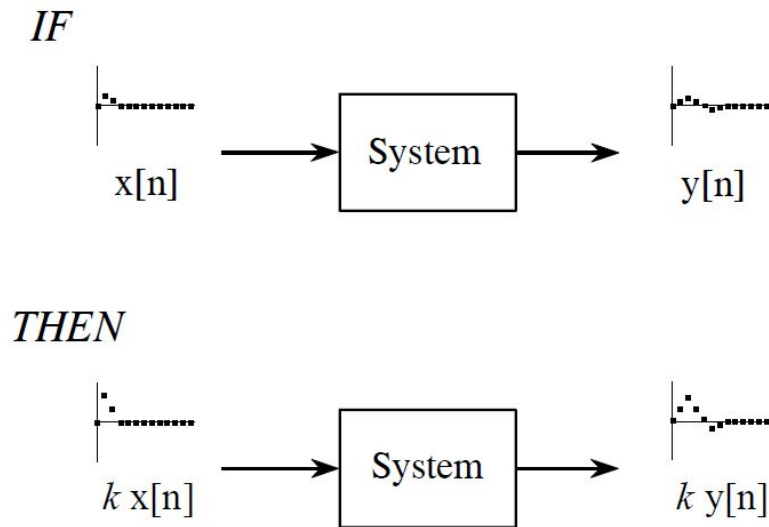


Figure: Illustration of homogeneity

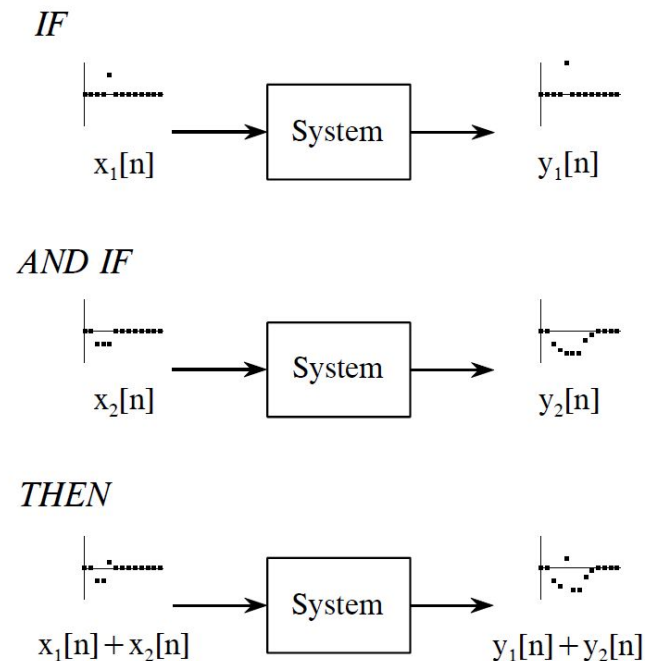


Figure: Illustration of additivity

Requirements of Linearity (contd.)

- **Shift invariance** is not a strict requirement but mandatory for most DSP techniques.

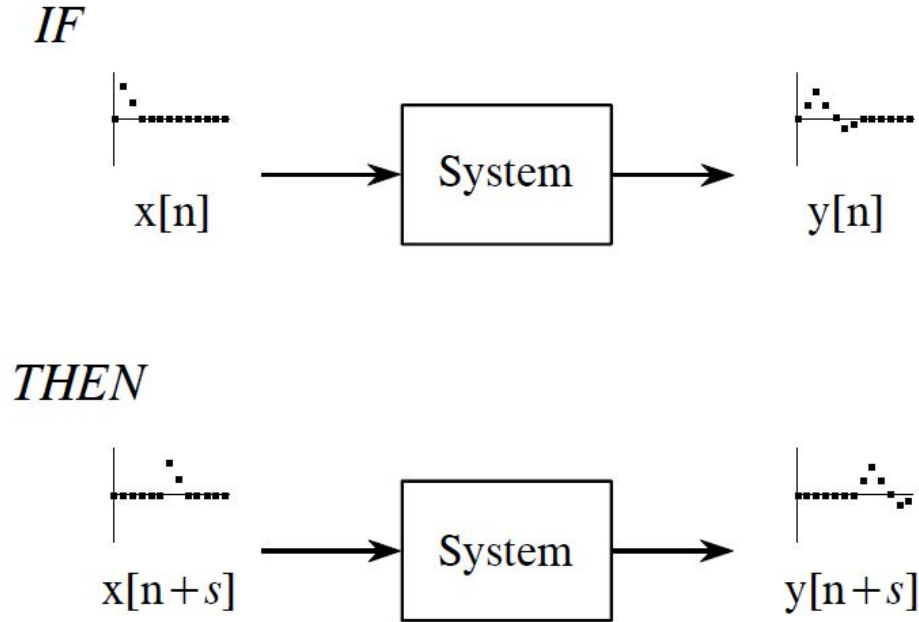


Figure: Illustration of shift invariance

Static Linearity and Sinusoidal Fidelity

- **Static Linearity**

- Defines how a linear system reacts when the signals aren't changing. For example, DC current.
- The static response of a linear system is the input multiplied by a constant

$$x_0(n) = 1 \longrightarrow y_0(n) = k$$

$$x_I(n) = 4 = 4 \cdot 1 = 4 x_0(n) \longrightarrow y_I(n) = 4 y_0(n) = 4k$$

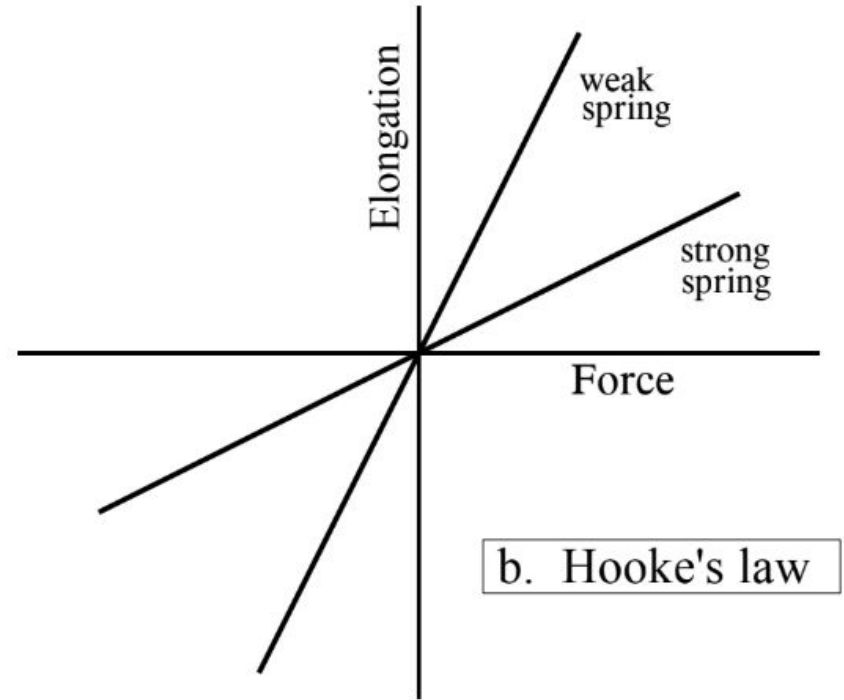
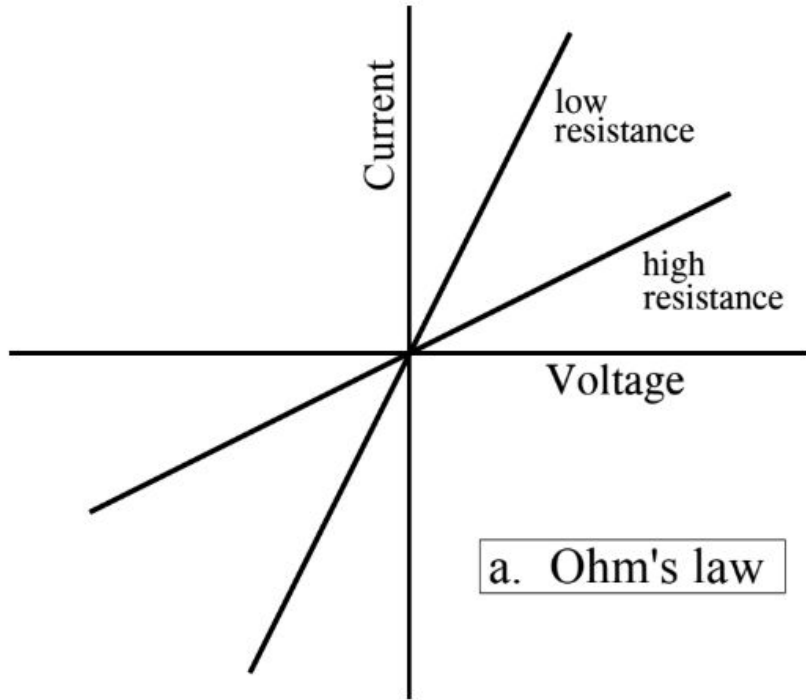


Figure: Two examples of static linearity. In (a), Ohm's law: the current through a resistor is equal to the voltage across the resistor divided by the resistance. In (b), Hooke's law: The elongation of a spring is equal to the applied force multiplied by the spring stiffness coefficient.

Static Linearity and Sinusoidal Fidelity (contd.)

- **Static Linearity (contd.)**

- All linear systems have the property of static linearity. The opposite isn't always true.
- However, if a system has static linearity and is memoryless, then the system **must be** linear.

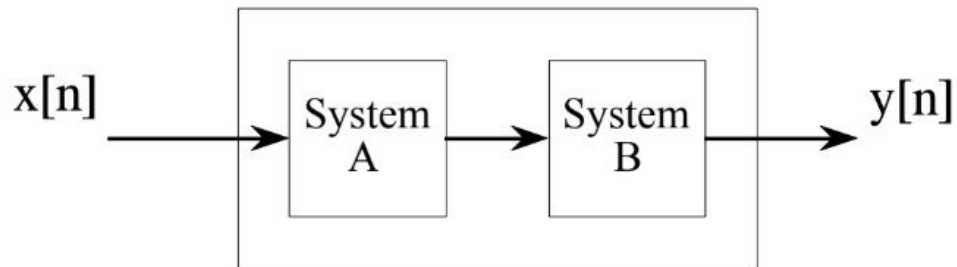
- **Sinusoidal Fidelity**

- If the input to a linear system is a sinusoidal wave, the output will also be a sinusoidal wave, and at exactly the same frequency as the input
- However, the *amplitude* and *phase* may be different
- Sinusoids are the only waveform to have this property
- The opposite isn't necessarily true

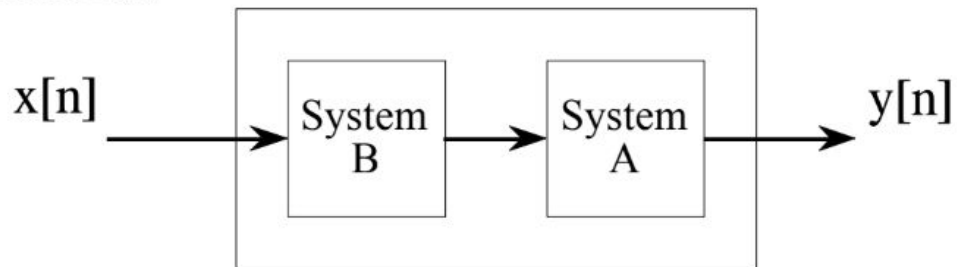
Special Properties of Linearity

- **Commutative:** Order of the systems in the cascade interconnection doesn't affect the overall output
- The overall combination after cascading some linear systems is also linear

IF

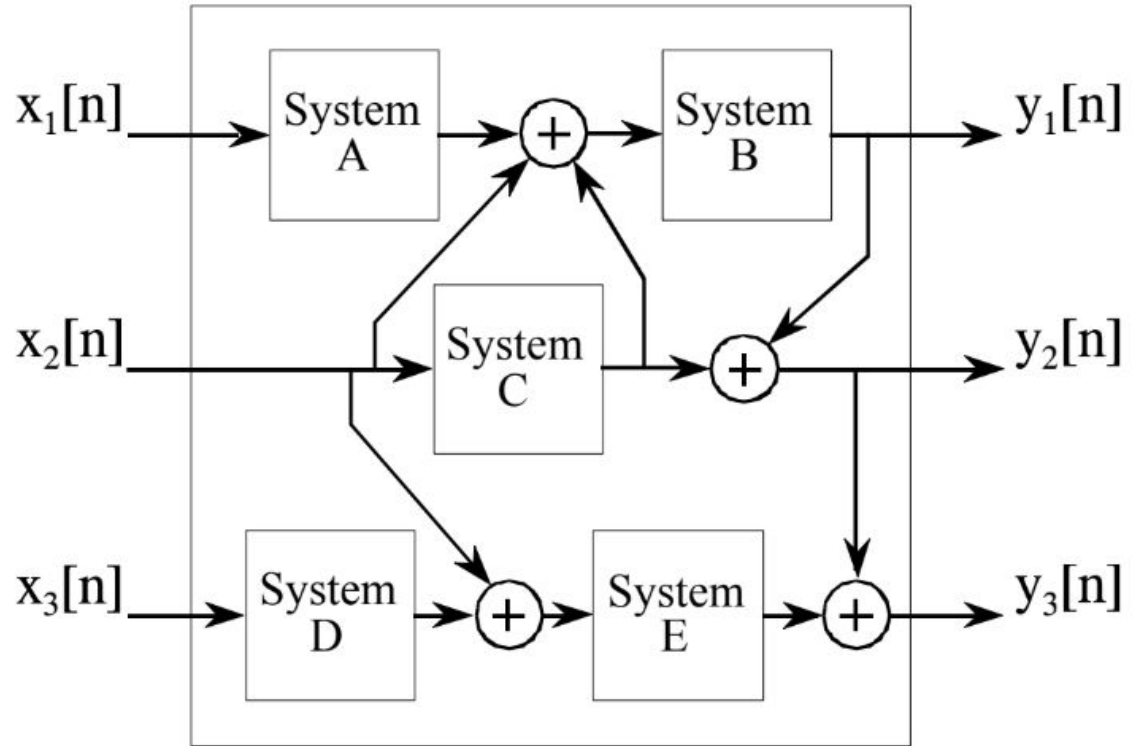


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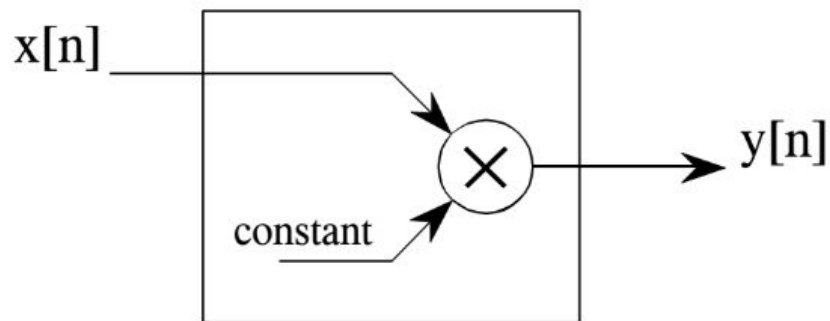
Special Properties of Linearity (contd.)

- **Superposition principle:** A system with multiple inputs and/or outputs will be linear if it is composed of linear subsystems and additions of signals



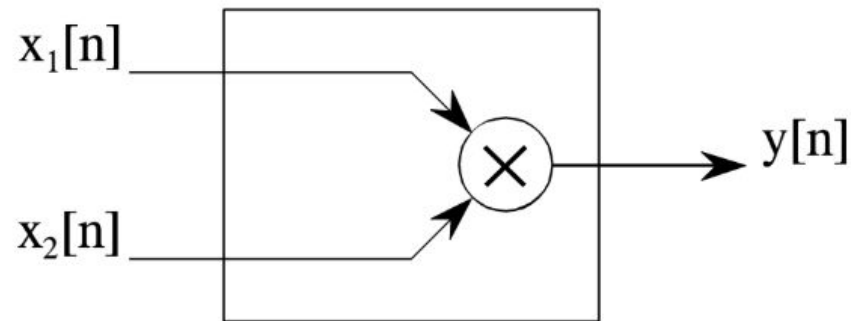
Special Properties of Linearity (contd.)

- Multiplicative? only for multiplications by constant values



Linear

a. Multiplication by a constant



Nonlinear

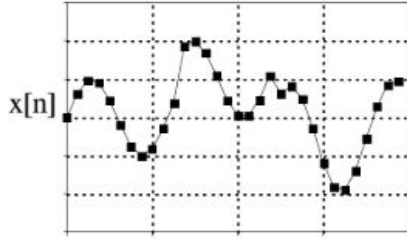
b. Multiplication of two signals

Figure: Linearity of multiplication. Multiplying a signal by a constant is a linear operation. In contrast, the multiplication of two signals is nonlinear.

Superposition: the Foundation of DSP

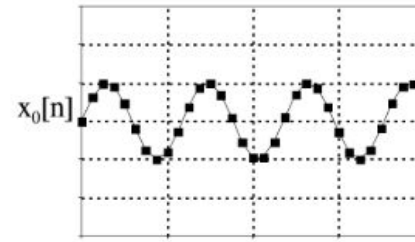
- **Synthesis:** Combining signals by *scaling* followed by *addition*
- **Decomposition**
 - Inverse operation of synthesis
 - A single signal is broken into multiple additive components
 - Can be infinite possible ways for any given signal

Superposition: the Foundation of DSP (contd.)

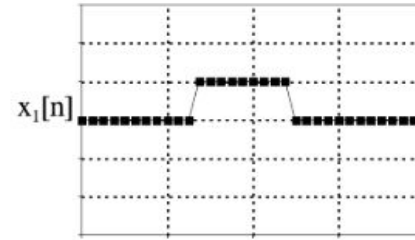


synthesis

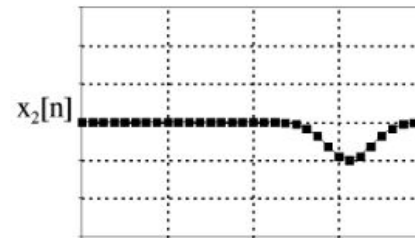
decomposition



+

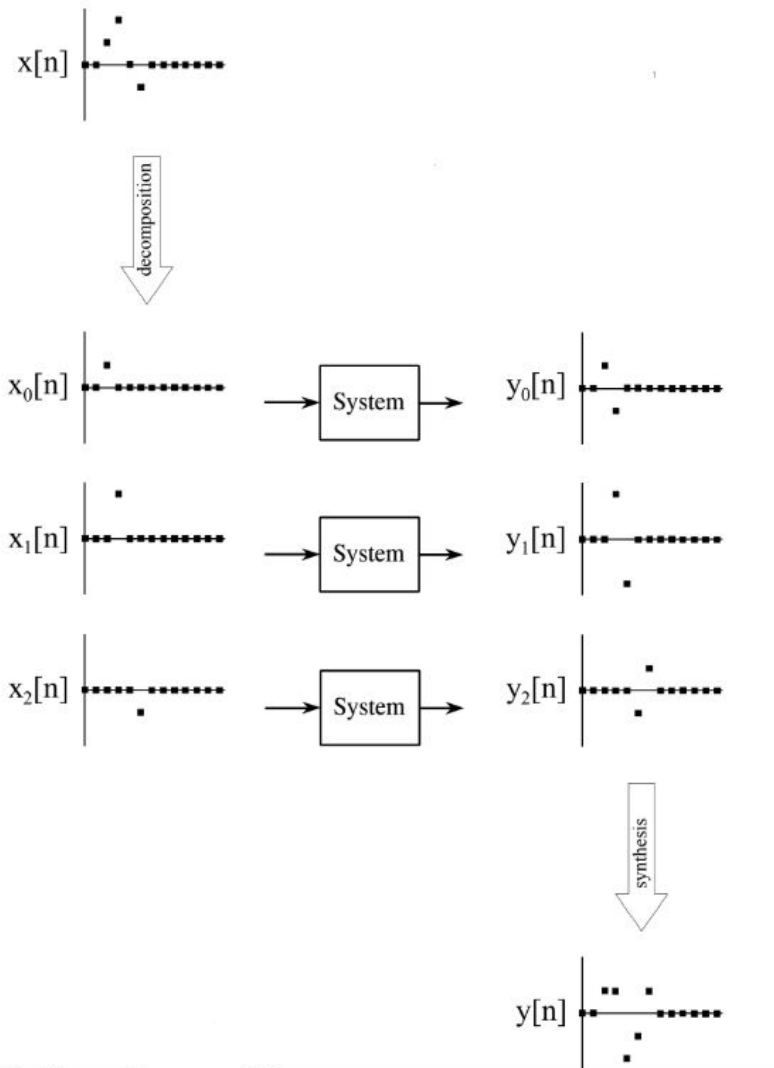


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Superposition: the Foundation of DSP (contd.)

- Instead of trying to understand how complicated signals are changed by a system, all we need to know is how simple (decomposed) signals are modified



Common Decompositions

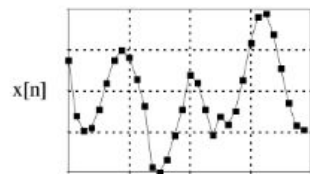
- **Impulse Decomposition**

- N samples signal $\longrightarrow$ N component signals containing N samples each
- Each component signal contains one point from the original signal (remainder of the values are zero) $\longrightarrow$ impulse signal
- Characterizes signals by looking at one point at a time
- Allows signals to be examined one sample at a time
- System is characterized by how it responds to an impulse, meaning the system's output can be calculated for any given input (convolution)

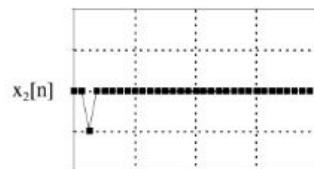
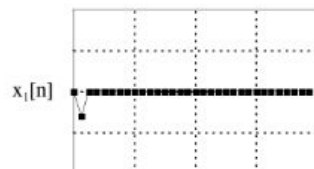
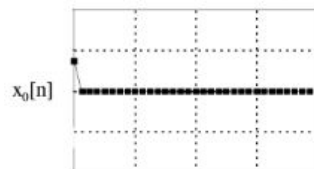
Common Decompositions (contd.)

- **Step Decomposition**

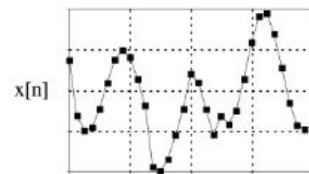
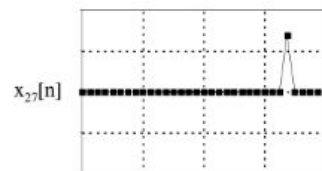
- N samples signal $\longrightarrow$ N component signals containing N samples each
- Each component signal is a step signal (the first samples are zero, the last samples are some constant value)
- The component signal is composed of zeros for points 0 through $k - 1$, while the remaining points have a value of: $x[k] - x[k - 1]$
- $x_0[n]$ has all of its samples equal to $x[0]$
- Characterizes signals by the *difference* between adjacent samples
- System is characterized by how they respond to a *change* in the input signal



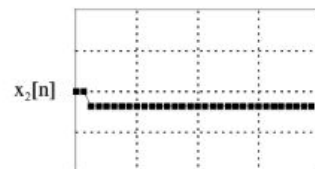
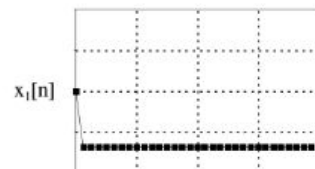
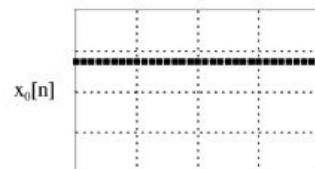
Impulse
Decomposition



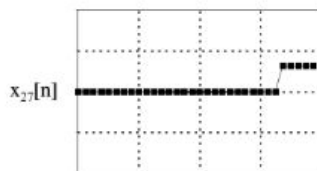
$\vdots$



Step
Decomposition



$\vdots$



Common Decompositions (contd.)

- **Even/Odd Decomposition**

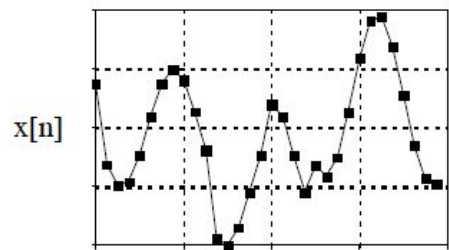
- N samples signal $\longrightarrow$ Two component signals containing N samples
- One component signal has **even symmetry**, other one has **odd symmetry**
- The **zeroth samples** in the **even and odd signals** are calculated as $x_E[0] = x[0]$ and $x_O[0] = 0$
- The indexes are considered running from 0 to $N-1$
- **Circular Symmetry!**


$$\begin{aligned} x_E[n] &= \frac{x[n] + x[N-n]}{2} \\ x_O[n] &= \frac{x[n] - x[N-n]}{2} \end{aligned}$$

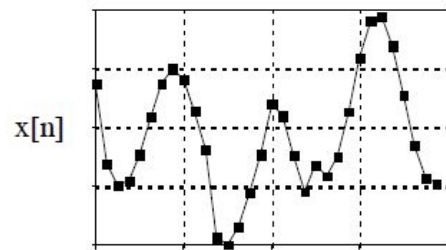
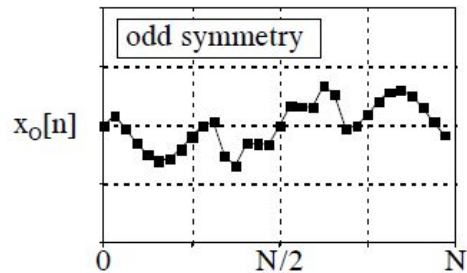
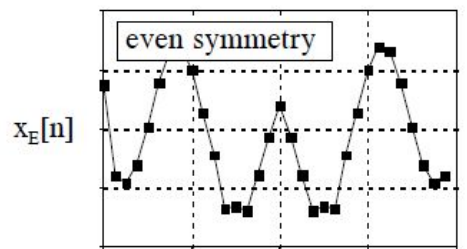
Common Decompositions (contd.)

- **Interlaced Decomposition**

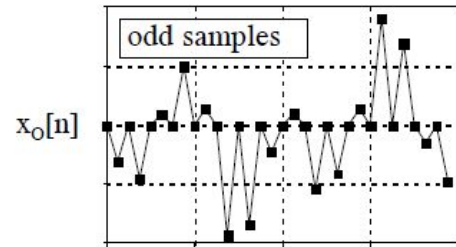
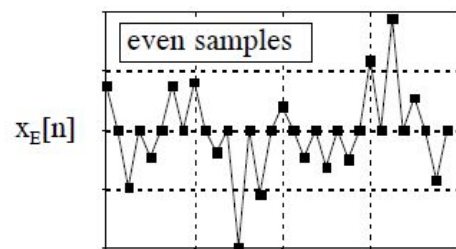
- N samples signal $\longrightarrow$ Two component signals containing N samples
- They are *even sample* signal and *odd sample* signal (not even and odd symmetry signals)
- Basis of Fast Fourier Transform (FFT) algorithm



Even/Odd
Decomposition



Interlaced
Decomposition

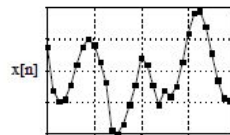


Common Decompositions (contd.)

- **Fourier Decomposition**

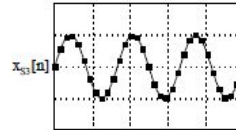
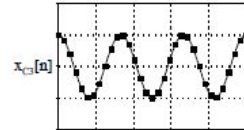
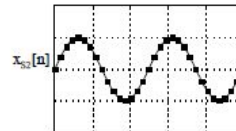
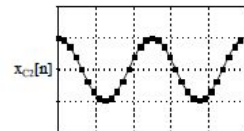
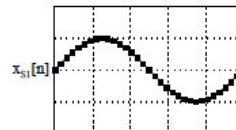
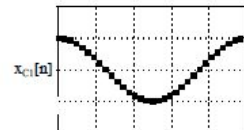
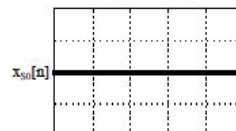
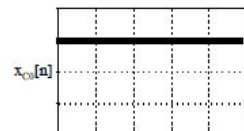
- N samples signal $\longrightarrow$ $N+2$ component signals
- Half of those signals are sine waves and half of them are cosine waves
- Why important?
 - A wide variety of signals are created from superimposed sinusoids. For example, audio signals.
 - Linear systems respond to sinusoids in a unique way: sinusoidal output signals for sinusoidal input signals
 - Basis of powerful techniques: *Fourier Analysis*, *Laplace*, *z-transforms* which subsequently are the basis of most cutting-edge DSP algorithms

Fourier
Decomposition



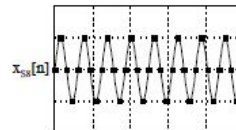
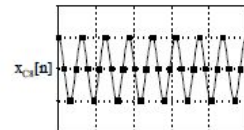
cosine waves

sine waves



⋮

⋮



Alternative to Linearity

- Only strategy for analyzing nonlinear systems is to make them resemble a linear system
- Three ways
 - Ignore the nonlinearity
 - Keep the signals very small
 - Apply a linearizing transform: *homomorphic* signal processing